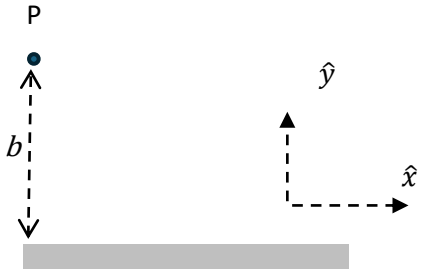
Physics 2 Problem Solving Video – Electric Field Part B – E field

1) A positively charged straight wire with a uniform linear charge density of λ, determine the electric field strength at point P, "b" meters above the wire and situated at the above the center of the wire of length L. Assume that a and L are constant and do not change in this problem.



A) Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.

B) Determine the incremental charge on the wire.

Answer: $dq = \lambda dx$

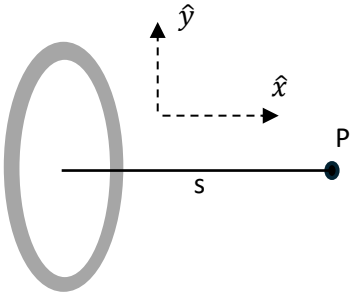
C) Determine the x component of the electric field strength.

Answer: $E_x = k\lambda\left[\frac{1}{(b^2+L^2)^{1/2}} - \frac{1}{b}\right]$

D) Determine the y component of the electric field strength.

Answer: $E_y = \frac{k\lambda}{b}\left[\frac{L}{(b^2+L^2)^{1/2}}\right]$

2) A circular ring with a radius R and a uniform linear charge density of λ is shown below, determine the electric field at point P along the x-axis which is at a distance of s from the center of the ring. Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.



Answer: : $E_x = \frac{k\lambda sR2\pi}{(b^2+L^2)^{3/2}}$